

Cost-Sensitive Customer Churn Prediction: A Machine Learning Pipeline for Telecom Retention

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Spring 2026

Abstract

This report presents a complete end-to-end machine learning pipeline for predicting customer churn in the telecommunications industry using the Telco Customer Churn dataset. The project encompasses exploratory data analysis, domain-driven feature engineering, class imbalance mitigation through SMOTE-based synthetic oversampling, comparative evaluation of nine classification architectures, and cost-sensitive decision threshold optimization under an asymmetric business utility function. Logistic Regression with SMOTE balancing emerged as the optimal model, achieving a maximum net profit of 20,040 monetary units at an optimal decision threshold $\tau^* = 0.32$. Post-hoc interpretability via SHAP feature attribution identified customer tenure, monthly charges, fiber optic service, month-to-month contracts, and electronic check payment as the five most influential churn drivers. The final model was deployed to generate test-set predictions for automated evaluation.

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1 Introduction

Customer attrition poses a significant financial threat to subscription-based business models. Telecommunications companies lose substantial recurring revenue when subscribers terminate their contracts, making proactive churn identification a critical operational priority. This project addresses the problem of constructing a highly calibrated classification model that predicts customer churn using demographic attributes, service subscription profiles, and account financial telemetry.

The project moves beyond traditional symmetric evaluation metrics by introducing a cost-sensitive framework that reflects real-world business trade-offs. Successfully identifying a churning customer yields a financial utility of +100 monetary units through targeted retention, while falsely flagging a loyal customer incurs a penalty of -20 units for unnecessary marketing outreach.

The full analysis code is available at: <https://github.com/fermow/SBU-Data-Science-2026>

2 Exploratory Data Analysis

2.1 Dataset Overview

The Telco Customer Churn dataset contains 7,043 subscriber records with 20 feature covariates and a binary target variable, `Churn`, indicating whether a customer terminated their contract within the observation window. The feature space spans three functional dimensions: demographic attributes (gender, SeniorCitizen, Partner, Dependents), service architecture (PhoneService, InternetService, OnlineSecurity, TechSupport, streaming services), and account financials (tenure, Contract type, PaymentMethod, MonthlyCharges, TotalCharges).

2.2 Data Quality and Cleaning

Structural inspection revealed that the `TotalCharges` column contained blank strings coerced into a generic text data type. After parsing to numeric, these entries became NaN values, which were imputed using the column median. No other missing values or structural anomalies were detected.

2.3 Target Distribution and Class Imbalance

The target variable exhibits pronounced class asymmetry. Non-churning customers constitute 73.5% of the dataset, while churners account for only 26.5%, yielding an approximate 2.77:1 ratio. This imbalance must be explicitly addressed during modeling to prevent classifiers from trivially predicting the majority class.

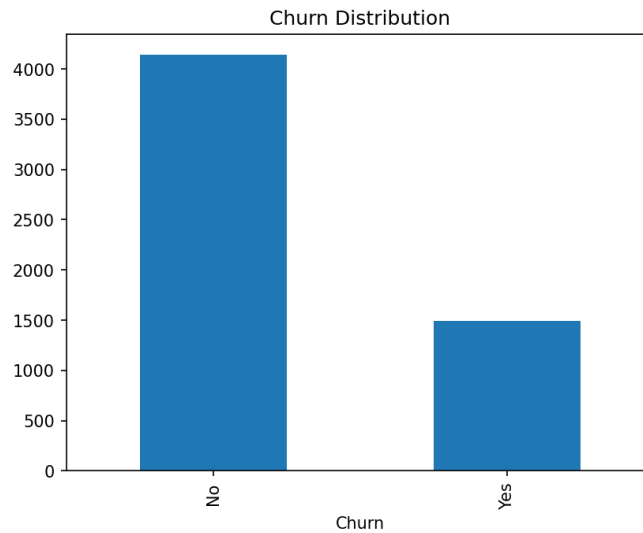


Figure 1: Distribution of the target variable showing class imbalance. Churners represent 26.5% of observations.

2.4 Bivariate Analysis

2.4.1 Contract Type

Contract duration exhibits the strongest univariate association with churn. Month-to-month contracts show a churn rate of 42.7%, compared to 11.1% for one-year contracts and only 2.9% for two-year contracts.

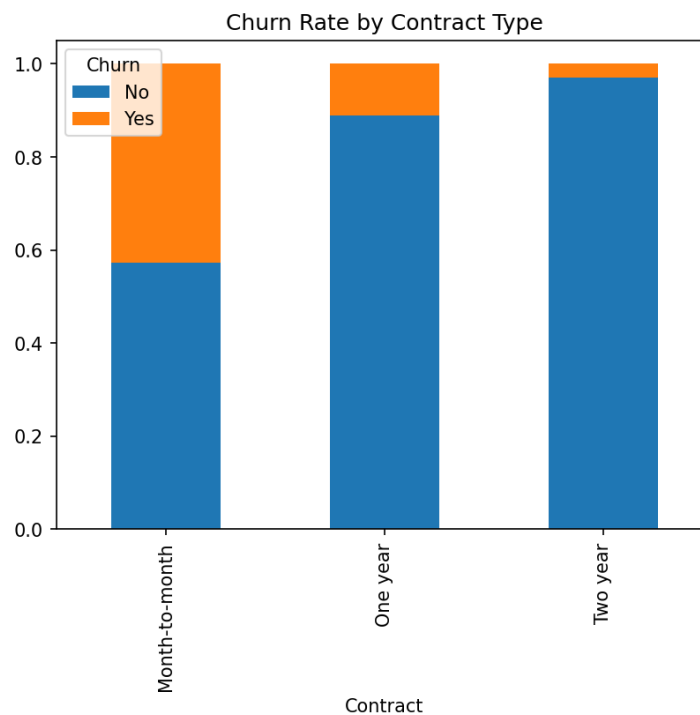


Figure 2: Churn rate by contract type. Month-to-month customers churn at dramatically higher rates.

2.4.2 Tenure

Customer tenure is inversely related to churn risk. Churners have a median tenure of 10 months compared to 38 months for retained customers, indicating that early-stage churn is the dominant pattern.

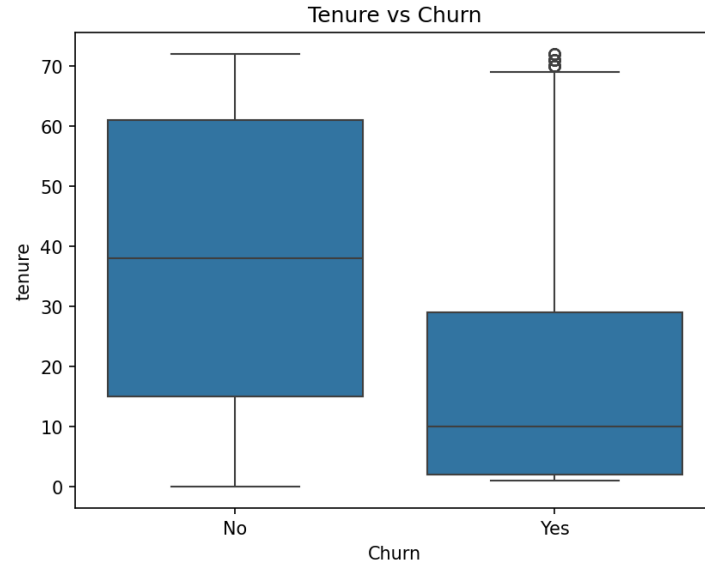


Figure 3: Tenure distribution by churn status. Churners exhibit significantly shorter tenure.

2.4.3 Internet Service and Payment Method

Fiber optic internet users churn at 42.1%, more than double the rate of DSL users (18.7%). Electronic check payment users churn at 45.7%, far exceeding automatic payment methods which average approximately 15%.

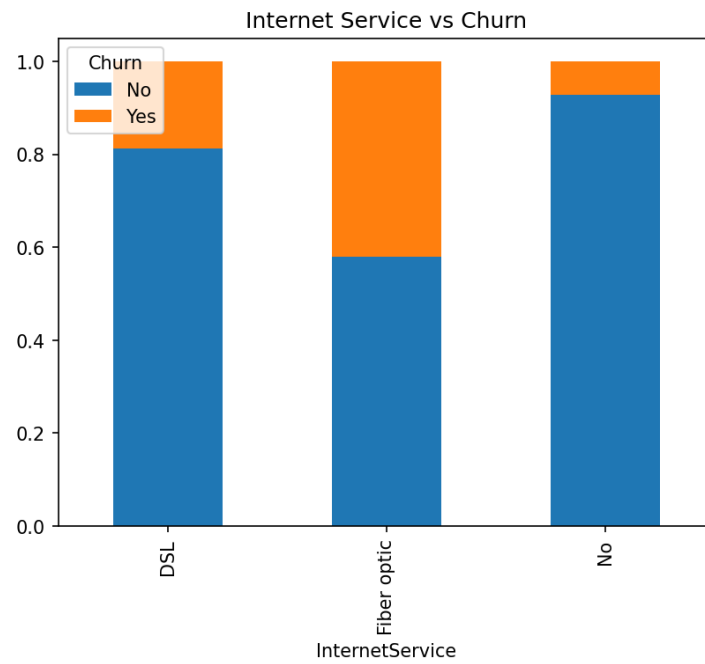


Figure 4: Internet service type versus churn rate. Fiber optic shows the highest attrition.

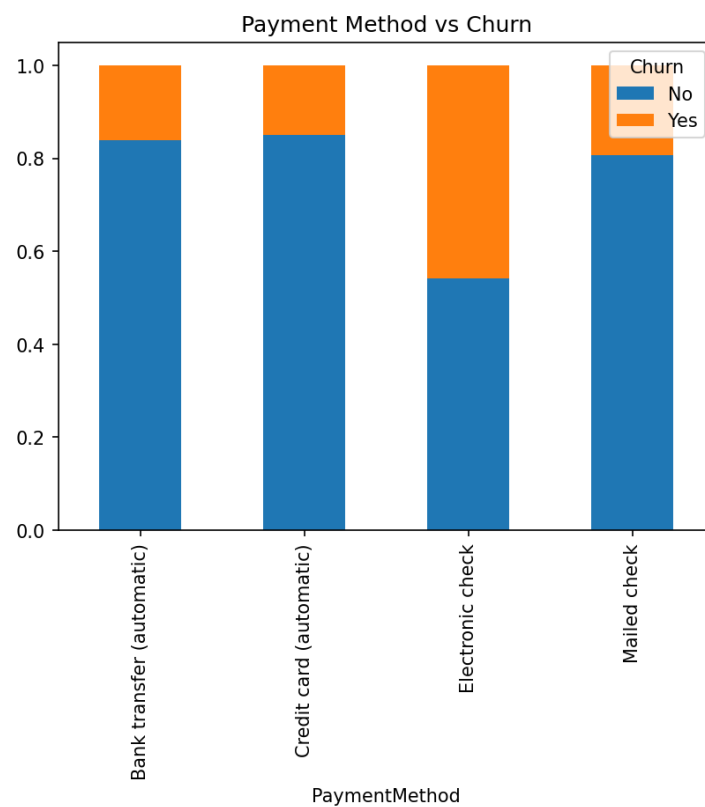


Figure 5: Payment method versus churn rate. Electronic check users churn at the highest rate.

2.5 Distributional Analysis

Kernel density estimation reveals that churners concentrate at lower tenure values and moderate-to-high monthly charges, while non-churners exhibit a bimodal tenure distribution with peaks at both short and long durations.

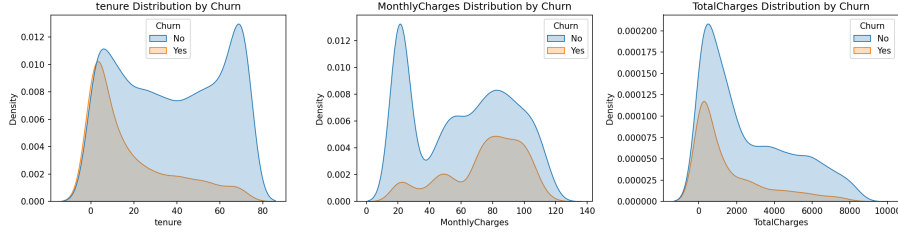


Figure 6: KDE plots of continuous features stratified by churn status.

2.6 Correlation Analysis

Both Pearson and Spearman correlation matrices were computed for numerical features. **TotalCharges** and **tenure** exhibit strong positive correlation ($r \approx 0.83$), indicating potential redundancy. **MonthlyCharges** shows moderate correlation with **TotalCharges**, while **SeniorCitizen** displays weak linear relationships with all other variables.

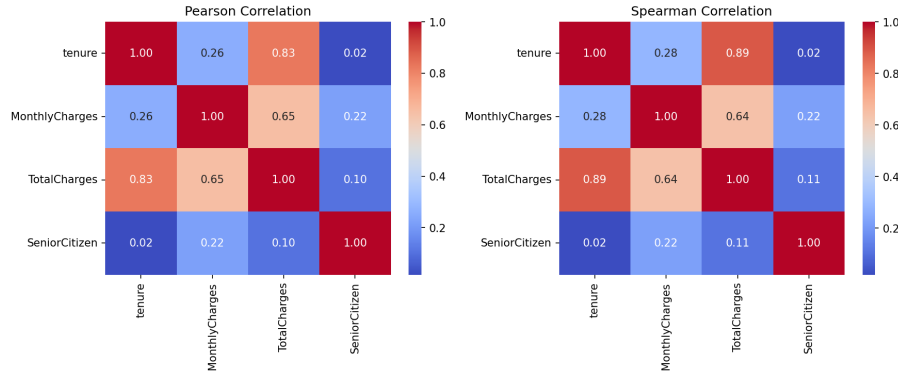


Figure 7: Pearson and Spearman correlation heatmaps for numerical features.

2.7 Analytical Synthesis

The EDA identifies four highly discriminative features: contract type, tenure, internet service type, and payment method. These variables exhibit clear separation between churn and non-churn populations and serve as primary candidates for feature engineering and modeling. The class imbalance ratio of 26.5% confirms the necessity of synthetic oversampling or cost-sensitive techniques.

3 Feature Engineering

Ten custom features were engineered to capture behavioral hypotheses that extend beyond raw covariates. Each feature encodes a specific business intuition about churn behavior.

Table 1: Engineered features and their behavioral hypotheses.

Feature	Type	Hypothesis
TenureRange	Ordinal	Customer lifecycle stage predicts retention behavior
TotalServices	Discrete	Service adoption depth signals engagement
Security	Binary	Security-conscious customers are more loyal
Entertainment	Binary	Streaming users have different churn patterns
SeniorTechSupport	Binary	Vulnerable seniors with support are more retained
BillingAndPayment	Binary	High-friction billing increases churn risk
MonthlyChargesRange	Ordinal	Pricing tiers have distinct churn profiles
AvgMonthlyPerService	Continuous	Cost efficiency per service affects retention
TenureContractRisk	Binary	Short tenure + month-to-month = maximum risk
LoyaltyFactor	Ordinal	Composite loyalty score [1–5] from contract + tenure

All categorical features were vectorized using OneHotEncoder for multi-class variables and binary encoding for dual-state variables. Numerical features were standardized using StandardScaler. All transformations were embedded within scikit-learn Pipeline objects to prevent data leakage during cross-validation.

4 Class Imbalance Mitigation

4.1 Baseline Assessment

A default Random Forest classifier was trained on the unweighted training partition to establish a baseline. The model achieved 79% overall accuracy but exhibited poor minority-class recall (47%), confirming that native training biases predictions toward the majority class.

4.2 Synthetic Oversampling

SMOTE (Synthetic Minority Oversampling Technique) was applied within the training folds to generate synthetic churning samples. This improved minority recall from 0.47 to 0.56 and F1-score from 0.54 to 0.59.

4.3 Cost-Penalized Learning and ADASYN

Class weighting via the `class_weight` parameter showed minimal improvement (recall 0.46). ADASYN, which focuses synthetic sampling on difficult regions, improved churn detection but introduced greater variance than SMOTE.

Table 2: Comparison of imbalance mitigation strategies on Random Forest.

Method	Recall	F1-score	Trade-off
Baseline (no mitigation)	0.47	0.54	Low churn detection
SMOTE	0.56	0.59	Best overall balance
ADASYN	0.54	0.57	Moderate improvement
Class weighting	0.46	0.54	Negligible effect

SMOTE was selected as the primary imbalance mitigation strategy for all subsequent models.

5 Multi-Model Evaluation

Nine classification architectures were implemented, trained on SMOTE-balanced data, and evaluated on the held-out validation set using stratified 80/20 splitting. Models requiring hyperparameter tuning were optimized via stratified 5-fold cross-validation.

5.1 Hyperparameter Search Spaces

Table 3: Hyperparameter search spaces for tuned models.

Model	Search Space
KNN	n_neighbors: [3, 5, 7, 9, 11], metric: [euclidean, manhattan]
Logistic Regression	C: [0.01, 0.1, 1, 10, 100]
Random Forest	n_estimators: [100, 200, 300, 500], max_depth: [None, 5, 10, 20, 30], min_samples_split: [2, 5, 10], min_samples_leaf: [1, 2, 4]
SVM	C: [0.1, 1, 10], kernel: [linear, rbf], gamma: [scale, auto]

5.2 Performance Comparison

Table 4: Validation performance across nine classification models (SMOTE-balanced).

Model	Accuracy	Precision	Recall	F1-score
Naive Bayes	0.70	0.46	0.83	0.59
Decision Tree	0.74	0.50	0.58	0.54
KNN (Tuned)	0.73	0.49	0.57	0.53
AdaBoost	0.78	0.59	0.56	0.57
Bagging	0.77	0.55	0.72	0.62
Logistic Regression	0.76	0.53	0.78	0.63
Random Forest	0.79	0.60	0.60	0.60
SVM (RBF)	0.76	0.53	0.60	0.56
MLP	0.75	0.55	0.58	0.56

Logistic Regression and Bagging achieved the highest churn recall (0.78 and 0.72 respectively), while Random Forest and AdaBoost delivered the best precision and overall stability.

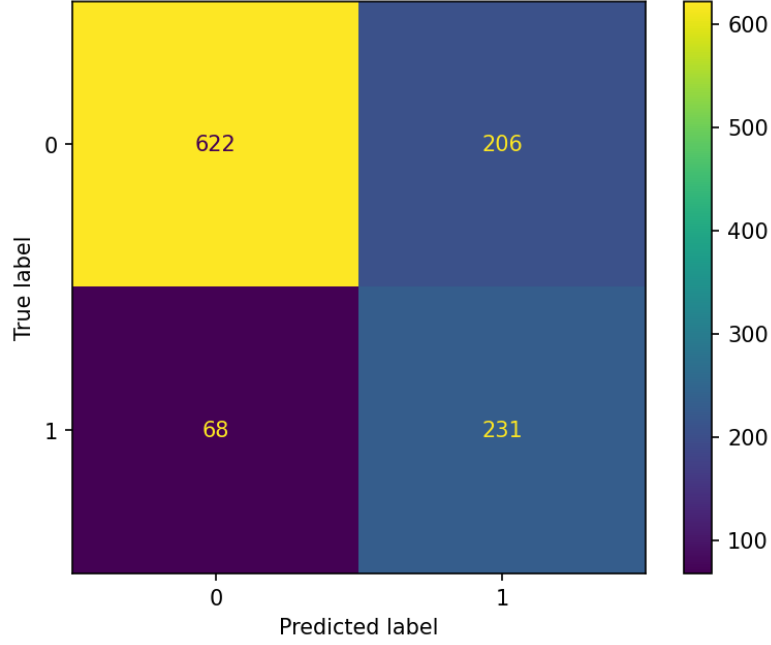


Figure 8: Confusion matrix for the top-performing Logistic Regression model.

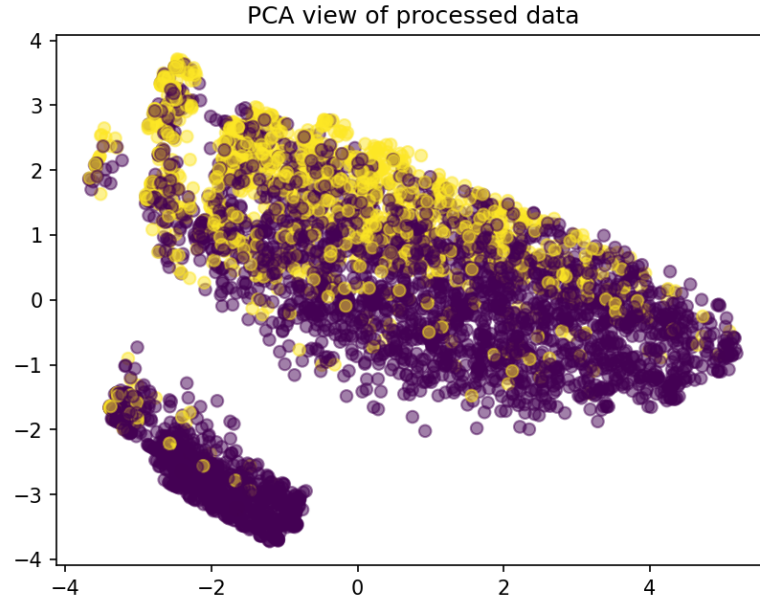


Figure 9: PCA visualization of the processed feature space, colored by churn status.

6 Business Utility Optimization

6.1 Asymmetric Cost Framework

Traditional accuracy-oriented metrics do not capture real-world business trade-offs. We define an asymmetric utility function:

$$U(\tau) = 100 \cdot \text{TP}(\tau) - 20 \cdot \text{FP}(\tau)$$

where TP and FP are functions of the decision threshold τ . The optimal threshold τ^* maximizes net profit.

6.2 Threshold Sweep Results

A systematic sweep across $\tau \in [0, 1]$ with step 0.01 was performed for each model.

Table 5: Optimal thresholds and maximum net profit by model.

Model	τ^*	Max Profit
Logistic Regression	0.32	20,040
AdaBoost	0.44	19,980
Random Forest	0.25	19,920
SVM	0.16	19,820
KNN	0.10	19,300
Bagging	0.13	19,200

Logistic Regression achieved the highest net profit of 20,040 monetary units at $\tau^* = 0.32$, demonstrating that a well-calibrated linear model can outperform more complex architectures under asymmetric cost structures.

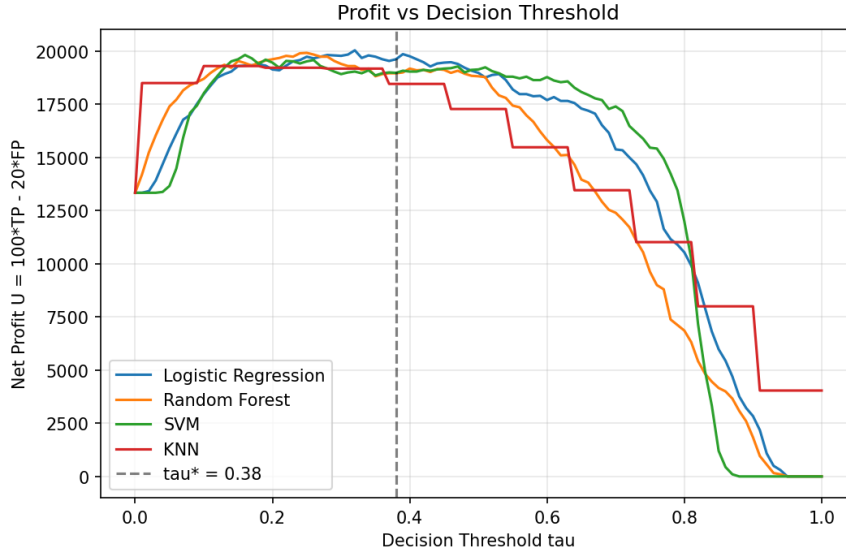


Figure 10: Net profit as a function of decision threshold for top models. The optimal threshold τ^* for Logistic Regression is marked.

6.3 Key Insight

Logistic Regression does not achieve the highest raw accuracy (0.76) or F1-score (0.63), yet it yields the maximum business profit. This occurs because its probability calibration – a natural property of the logistic sigmoid – produces well-calibrated predictions that allow the threshold sweep to precisely balance true positives and false positives. In contrast, models with higher precision or accuracy may have less calibrated probability estimates, making it harder to find an optimal operating point.

7 Model Interpretability

7.1 SHAP Feature Attribution

SHAP (SHapley Additive exPlanations) was applied to the optimal Logistic Regression model to quantify per-feature contributions to individual predictions.

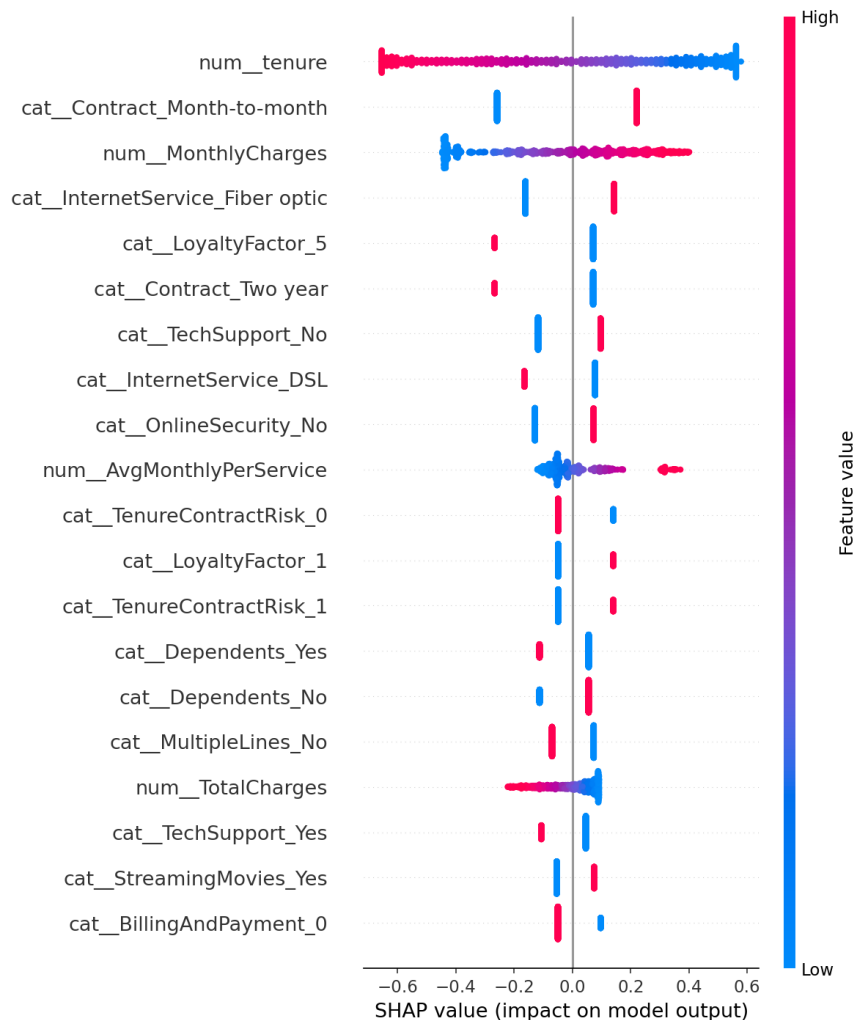


Figure 11: SHAP summary beeswarm plot for the Logistic Regression model. Features are ranked by mean absolute impact.

7.2 Top Five Drivers and EDA Connections

1. **tenure** (Most Impactful): Lower tenure (blue) increases churn probability; higher tenure (red) decreases it. This aligns with the EDA finding that churners have a median tenure of 10 months versus 38 months for loyal customers.
2. **MonthlyCharges**: Lower charges reduce churn risk; higher charges increase it. EDA revealed churners average \$74.86/month versus \$61.34 for non-churners.
3. **InternetService_Fiber optic**: Fiber optic subscription strongly increases churn probability. EDA showed a 42.1% churn rate for fiber optic users versus 18.7% for DSL.

4. **Contract_Month-to-month:** Month-to-month contracts are a dominant churn catalyst. EDA confirmed 42.7% churn for this group versus 2.9% for two-year contracts.
5. **PaymentMethod_Electronic check:** Electronic check payment significantly raises churn risk. EDA found 45.7% churn for electronic check users versus approximately 15% for automatic methods.

The SHAP attributions directly validate the EDA findings and confirm that the model has learned the same discriminative patterns observed during exploratory analysis.

8 Conclusion

This project successfully constructed a complete cost-sensitive churn prediction pipeline for the telecommunications domain. Key findings include:

1. **EDA identified four highly discriminative predictors:** contract type, tenure, internet service type, and payment method.
2. **SMOTE oversampling** improved minority recall from 0.47 to 0.56 relative to the unweighted baseline.
3. **Logistic Regression with SMOTE** achieved the highest business utility (20,040 monetary units) at $\tau^* = 0.32$, outperforming more complex architectures due to superior probability calibration.
4. **SHAP attribution** confirmed that tenure, monthly charges, fiber optic service, month-to-month contracts, and electronic check payment are the dominant churn drivers, directly connected to empirical EDA trends.

The final model was deployed on the test set using the empirically determined optimal threshold, generating predictions for the automated evaluation system.

A Appendix

A.1 Technology Stack

- Python 3.x
- pandas, numpy (data manipulation)
- scikit-learn (pipelines, preprocessing, models, cross-validation)
- imbalanced-learn (SMOTE, ADASYN)
- shap (model interpretability)
- matplotlib, seaborn (visualization)

A.2 Data Preprocessing Pipeline

All transformations were embedded in scikit-learn `Pipeline` objects:

- `ColumnTransformer` with `OneHotEncoder` for categorical features and `StandardScaler` for numerical features
- `ImbPipeline` for SMOTE-integrated pipelines ensuring oversampling occurs only within training folds

A.3 Reproducibility

All random state parameters across train-test splits, synthetic samplers, and estimators were fixed to `random_state=42` to ensure complete reproducibility.

A.4 Code Repository

The complete analysis is available in the accompanying Jupyter notebooks hosted on GitHub: <https://github.com/fermow/SBU-Data-Science-2026>